

04 Main Analyses + Education

The ‘Immigration Issue’ Misnomer in Human Capital Research:
Intersectional Inequalities in Adult Education and Literacy in Upper- and
Middle-Income Countries

Author Omitted for Blind Peer Review

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Setup

```
knitr::opts_chunk$set(warning = FALSE, message = FALSE)
```

```
packages <- c('tidyverse',  
              'readr',  
              'knitr',  
              'here',  
              'ggplot2',  
              'ggrepel',  
              'ragg',  
              'kableExtra',
```

```
'countrycode',
'readxl',
'jtools',
'lme4',
'lmerTest',
'broom.mixed',
'viridis',
'survey',
'ggpubr',
'purrr')
```

```
pacman::p_load(packages, character.only = T)
```

Data

```
df2 <- read_rds(here("Data", "df2.RDS"))
```

Table A5. GLM/Multilevel

Disclosure: I first tried running with completed primary only (ed3 == 1) as the intercategory, but the case numbers were too small for some of the categories and prediction was unstable. Then I switched to completed secondary versus completed tertiary.

Data Prep & Centering

```
# Create education dummies and filter full valid sample
```

```
df2_sub <- df2 %>%
  mutate(
    ed3_secondary = as.integer(ed3 == 2 | ed3 == 1),
    ed3_tertiary = as.integer(ed3 == 3)
  ) %>%
  subset(ed_currently != 1)
```

```
# Within/Between decomposition for the FULL sample
```

```
df2_sub <- df2_sub %>%
  group_by(iso3c) %>%
  mutate(
    male = ifelse(female == 1, 0, 1),
    NFE12_between = sum(SPFWT0 * NFE12, na.rm = TRUE) /
sum(SPFWT0[!is.na(NFE12)], na.rm = TRUE),
    female_between = sum(SPFWT0 * female, na.rm = TRUE) /
sum(SPFWT0[!is.na(female)], na.rm = TRUE),
    male_between = sum(SPFWT0 * male, na.rm = T) / sum(SPFWT0[!is.na(male)],
na.rm = TRUE),
    children_between = sum(SPFWT0 * children, na.rm = TRUE) /
sum(SPFWT0[!is.na(children)], na.rm = TRUE),
    ed3_secondary_between = sum(SPFWT0 * ed3_secondary, na.rm = TRUE) /
sum(SPFWT0[!is.na(ed3_secondary)], na.rm = TRUE),
    ed3_tertiary_between = sum(SPFWT0 * ed3_tertiary, na.rm = TRUE) /
sum(SPFWT0[!is.na(ed3_tertiary)], na.rm = TRUE),
    forborn_between = sum(SPFWT0 * FORBORN, na.rm = TRUE) /
```

```

sum(SPFWT0[!is.na(FORBORN)], na.rm = TRUE),
  forlang_between = sum(SPFWT0 * FORLANG, na.rm = TRUE) /
sum(SPFWT0[!is.na(FORLANG)], na.rm = TRUE),

  NFE12_within = NFE12 - NFE12_between,
  children_within = children - children_between,
  female_within = female - female_between,
  ed3_secondary_within = ed3_secondary - ed3_secondary_between,
  ed3_tertiary_within = ed3_tertiary - ed3_tertiary_between,
  forborn_within = FORBORN - forborn_between,
  forlang_within = FORLANG - forlang_between
) %>%
ungroup()

# Build country-level z-scored between effects
between_vars <- c("NFE12_between", "female_between", "children_between",
  "ed3_secondary_between", "ed3_tertiary_between",
  "forborn_between", "forlang_between")

country_between_scaled <- function(dat, between_vars, suffix = "_z") {
  dat %>%
    distinct(iso3c, across(all_of(between_vars))) %>%
    mutate(across(all_of(between_vars), ~{
      m <- mean(.x, na.rm = TRUE)
      s <- sd(.x, na.rm = TRUE)
      if (is.na(s) || s == 0) NA_real_ else as.numeric((.x - m) / s)
    }, .names = paste0("{.col}", suffix))) %>%
    select(iso3c, ends_with(suffix))
}

df2_sub <- df2_sub %>% left_join(country_between_scaled(df2_sub,
  between_vars), by = "iso3c")

### The Single Pooled GLM

m4_pooled <- lmer(
  skill_reading ~
    # Base fixed effects
    NFE12_within + NFE12_between_z +
    female_within + female_between_z +
    children_within + children_between_z +
    forlang_within + forlang_between_z +
    forborn_within + forborn_between_z +
    #ed3_secondary_within + ed3_secondary_between_z + # this can't be in the
model because of collinearity
    ed3_tertiary_within + ed3_tertiary_between_z +
    gdppc_k_log +
    #Intercategorical approach
    female_within * forborn_within * children_within * forlang_within *
    NFE12_within * ed3_tertiary_within +

```

```
    (1 | iso3c),  
    data = df2_sub  
)  
  
# Export summary  
export_summs(m4_pooled, statistics = "all", error_pos = "right", error_format  
= "{std.error}",  
             to.file = "docx", file.name = here("Results", "TS1.docx"))
```

	Model 1	
(Intercept)	207.81 **	(60.09)
NFE12_within	18.35 ***	(0.31)
NFE12_between_z	11.64 **	(3.79)
female_within	-0.24	(0.29)
female_between_z	-6.00	(2.98)
children_within	-8.82 ***	(0.32)
children_between_z	-6.02	(3.51)
forlang_within	-17.94 ***	(0.66)
forlang_between_z	-2.27	(3.99)
forborn_within	-17.18 ***	(0.51)
forborn_between_z	-3.94	(3.93)
ed3_tertiary_within	37.39 ***	(0.30)
ed3_tertiary_between_z	-1.07	(3.25)
gdppc_k_log	14.31	(16.56)
female_within:forborn_within	-0.39	(1.01)
female_within:children_within	-3.03 ***	(0.63)
children_within:forborn_within	-2.78 *	(1.10)
female_within:forlang_within	3.11 *	(1.29)
forlang_within:forborn_within	-3.88 ***	(1.12)
children_within:forlang_within	-0.85	(1.36)
NFE12_within:female_within	-1.91 **	(0.63)
NFE12_within:forborn_within	3.18 **	(1.06)
NFE12_within:children_within	-3.44 ***	(0.68)
NFE12_within:forlang_within	0.56	(1.34)
female_within:ed3_tertiary_within	-4.20 ***	(0.61)
forborn_within:ed3_tertiary_within	-2.25 *	(1.04)
children_within:ed3_tertiary_within	2.01 **	(0.65)
forlang_within:ed3_tertiary_within	4.35 **	(1.33)
NFE12_within:ed3_tertiary_within	-6.20 ***	(0.63)
female_within:children_within:forborn_within	-0.36	(2.21)
female_within:forlang_within:forborn_within	2.17	(2.07)
female_within:children_within:forlang_within	2.57	(2.72)
children_within:forlang_within:forborn_within	-3.50	(2.21)
NFE12_within:female_within:forborn_within	1.42	(2.12)

NFE12_within:female_within:children_within	-0.82	(1.35)
NFE12_within:children_within:forborn_within	1.53	(2.32)
NFE12_within:female_within:forlang_within	0.71	(2.67)
NFE12_within:forlang_within:forborn_within	6.48 **	(2.19)
NFE12_within:children_within:forlang_within	-3.97	(2.80)
female_within:forborn_within:ed3_tertiary_within	-1.59	(2.07)
female_within:children_within:ed3_tertiary_within	1.99	(1.30)
children_within:forborn_within:ed3_tertiary_within	2.10	(2.26)
female_within:forlang_within:ed3_tertiary_within	1.59	(2.65)
forlang_within:forborn_within:ed3_tertiary_within	4.33 *	(2.16)
children_within:forlang_within:ed3_tertiary_within	4.53	(2.78)
NFE12_within:female_within:ed3_tertiary_within	-1.60	(1.25)
NFE12_within:forborn_within:ed3_tertiary_within	2.28	(2.12)
NFE12_within:children_within:ed3_tertiary_within	3.57 **	(1.34)
NFE12_within:forlang_within:ed3_tertiary_within	5.89 *	(2.69)
female_within:children_within:forlang_within:forborn_within	7.69	(4.41)
NFE12_within:female_within:children_within:forborn_within	0.13	(4.65)
NFE12_within:female_within:forlang_within:forborn_within	-3.32	(4.38)
NFE12_within:female_within:children_within:forlang_within	-5.63	(5.60)
NFE12_within:children_within:forlang_within:forborn_within	9.69 *	(4.65)
female_within:children_within:forborn_within:ed3_tertiary_within	3.35	(4.52)
female_within:forlang_within:forborn_within:ed3_tertiary_within	-0.86	(4.31)
female_within:children_within:forlang_within:ed3_tertiary_within	-0.57	(5.57)
children_within:forlang_within:forborn_within:ed3_tertiary_within	-0.78	(4.58)
NFE12_within:female_within:forborn_within:ed3_tertiary_within	1.12	(4.25)
NFE12_within:female_within:children_within:ed3_tertiary_within	-6.77 *	(2.69)
NFE12_within:children_within:forborn_within:ed3_tertiary_within	6.65	(4.63)
NFE12_within:female_within:forlang_within:ed3_tertiary_within	-8.54	(5.37)
NFE12_within:forlang_within:forborn_within:ed3_tertiary_within	-2.48	(4.42)
NFE12_within:children_within:forlang_within:ed3_tertiary_within	1.66	(5.63)
NFE12_within:female_within:children_within:forlang_within:forborn_within	15.84	(9.30)
female_within:children_within:forlang_within:forborn_within:ed3_tertiary_within	-3.97	(9.16)
NFE12_within:female_within:children_within:forborn_within:ed3_tertiary_within	-11.34	(9.26)
NFE12_within:female_within:forlang_within:forborn_within:ed3_tertiary_within	21.31 *	(8.84)
NFE12_within:female_within:children_within:forlang_within:ed3_tertiary_within	-14.51	(11.25)
NFE12_within:children_within:forlang_within:forborn_within:ed3_tertiary_within	-5.80	(9.38)
NFE12_within:female_within:children_within:forlang_within:forborn_within:ed3_tertiary_within	14.30	(18.77)

nobs	122641
nobs.1	122641.00
sigma	43.68
logLik	-637165.56
AIC	1274477.13
BIC	1275186.47
REMLcrit	1274331.13
df.residual	122568.00
p.value	
r.squared	0.35
r.squared.fixed	0.30
group.nobs.iso3c	28.00

*** p < 0.001; ** p < 0.01; * p < 0.05.

Predicted ATEs

```
# Filter for complete cases for prediction baseline
df2r <- subset(df2_sub, !is.na(female) & !is.na(NFE12) &
!is.na(children_within) &
!is.na(forborn_within) & !is.na(forlang_within) &
!is.na(ed3) &
!is.na(gdppc) & !is.na(skill_reading))

# Grand means for recentering predictions
gm_female <- mean(df2r$female_between, na.rm = TRUE)
gm_forborn <- mean(df2r$forborn_between, na.rm = TRUE)
gm_children <- mean(df2r$children_between, na.rm = TRUE)
gm_forlang <- mean(df2r$forlang_between, na.rm = TRUE)
gm_nfe12 <- mean(df2r$NFE12_between, na.rm = TRUE)
gm_ed3_sec <- mean(df2r$ed3_secondary_between, na.rm = TRUE)
gm_ed3_tert <- mean(df2r$ed3_tertiary_between, na.rm = TRUE)

# Function to calculate ATE for a specific profile
calc_ate <- function(is_female, is_forborn, is_child, is_forlang,
is_tertiary) {

  # Set secondary to 1 to predict the non-tertiary reference group
  is_secondary <- ifelse(is_tertiary == 1, 0, 1)

  base_df <- df2r %>% mutate(
    female_within = is_female - gm_female,
    forborn_within = is_forborn - gm_forborn,
    children_within = is_child - gm_children,
    forlang_within = is_forlang - gm_forlang,
    ed3_tertiary_within = is_tertiary - gm_ed3_tert,
    ed3_secondary_within = is_secondary - gm_ed3_sec
  )

  df_0 <- base_df %>% mutate(NFE12_within = 0 - gm_nfe12)
  df_1 <- base_df %>% mutate(NFE12_within = 1 - gm_nfe12)

  pred0 <- predict(m4_pooled, newdata = df_0, re.form = NA)
  pred1 <- predict(m4_pooled, newdata = df_1, re.form = NA)

  tt <- t.test(pred1, pred0, paired = FALSE)

  tibble(
    AME = mean(pred1 - pred0),
    lower = tt$conf.int[1],
    upper = tt$conf.int[2]
  )
}
```

```

# Define the 12 Intersectional Profiles
profiles <- tribble(
  ~type_id, ~type_label, ~is_female, ~is_forborn, ~is_child, ~is_forlang,
  "1f", "1f. Foreign-born mother\nnon-native speaker", 1, 1, 1, 1,
  "2f", "2f. Foreign-born no children female\nnon-native speaker", 1, 1, 0,
1,
  "3f", "3f. Foreign-born mother\nnative speaker", 1, 1, 1, 0,
  "4f", "4f. Foreign-born no children female\nnative speaker", 1, 1, 0, 0,
  "5f", "5f. Native-born mother\nnative speaker", 1, 0, 1, 0,
  "6f", "6f. Native-born no children female\nnative speaker", 1, 0, 0, 0,

  "1m", "1m. Foreign-born father\nnon-native speaker", 0, 1, 1, 1,
  "2m", "2m. Foreign-born no children male\nnon-native speaker", 0, 1, 0, 1,
  "3m", "3m. Foreign-born father\nnative speaker", 0, 1, 1, 0,
  "4m", "4m. Foreign-born no children male\nnative speaker", 0, 1, 0, 0,
  "5m", "5m. Native-born father\nnative speaker", 0, 0, 1, 0,
  "6m", "6m. Native-born no children male\nnative speaker", 0, 0, 0, 0
)

# Cross profiles with education levels and calculate ATEs
ame_all <- crossing(
  profiles,
  is_tertiary = c(0, 1)
) %>%
  mutate(
    res = pmap(list(is_female, is_forborn, is_child, is_forlang,
is_tertiary), calc_ate)
  ) %>%
  unnest(res)

```

Prepare Data for Plotting

```

ame_all <- ame_all %>%
  mutate(
    ed_label = ifelse(is_tertiary == 1, "Tertiary", "Non-Tertiary"),
    idx = as.integer(substr(type_id, 1, 1)),
    sex = substr(type_id, 2, 2)
  )

# Set factor order identical to original layout
type_order <- ame_all %>% distinct(type_label, idx, sex) %>% arrange(idx,
sex) %>% pull(type_label)
ame_all$type_label <- factor(ame_all$type_label, levels = rev(type_order))

# Add binary dot indicators
ame_all <- ame_all %>%
  mutate(
    Female = ifelse(sex == "f", 1, 0),
    `Foreign-born` = ifelse(is_forborn == 1, 1, 0),
    `Non-native-speaker` = ifelse(is_forlang == 1, 1, 0),

```

```

    Parent = ifelse(is_child == 1, 1, 0)
  )

pal12 <- viridis(n = nlevels(ame_all$type_label), option = "D", begin = 0.1,
end = 0.9)

```

Generate Main Plots' Components

```

dot_df <- ame_all %>%
  filter(is_tertiary == 0) %>% # Only need one set of dots for the labels
  select(type_label, Female, `Foreign-born`, `Non-native-speaker`, Parent)
%>%
  pivot_longer(cols = c(Female, `Foreign-born`, `Non-native-speaker`,
Parent), names_to = "variable", values_to = "value") %>%
  mutate(variable = factor(variable, levels = c("Female", "Foreign-born",
"Non-native-speaker", "Parent")))

p_dots <- ggplot(dot_df, aes(x = variable, y = type_label)) +
  geom_point(data = subset(dot_df, value == 0), shape = 21, size = 3.2,
stroke = 0.5, fill = "white", color = "black") +
  geom_point(data = subset(dot_df, value == 1), shape = 16, size = 3.2, color
= "black") +
  labs(x = NULL, y = NULL) + theme_minimal(base_size = 12) +
  theme(axis.text.y = element_blank(), axis.ticks.y = element_blank(),
        panel.grid.major.y = element_blank(), panel.grid.minor =
element_blank(), panel.grid.major.x = element_blank(),
        axis.text.x = element_text(size = 10, angle = 45, hjust = 1))

# Non-Tertiary Plot
p_bar <- ggplot(subset(ame_all, is_tertiary == 0), aes(x = type_label, y =
AME, fill = type_label)) +
  geom_col(width = 0.55, color = "black") +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.15) +
  scale_fill_manual(values = pal12) +
  scale_y_continuous(breaks = seq(-10, 40, by = 5)) +
  coord_flip(ylim = c(-1, 35)) +
  labs(x = NULL, y = "Non-Tertiary\nEducation") + theme_minimal(base_size =
12) +
  theme(axis.text.y = element_blank(), axis.ticks.y = element_blank(),
        panel.grid.major.y = element_blank(), panel.grid.minor =
element_blank(), legend.position = "none")

# Tertiary Plot
p_bar_t <- ggplot(subset(ame_all, is_tertiary == 1), aes(x = type_label, y =
AME, fill = type_label)) +
  geom_col(width = 0.55, color = "black") +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.15) +
  scale_fill_manual(values = pal12) +
  scale_y_continuous(breaks = seq(-10, 40, by = 5)) +
  coord_flip(ylim = c(-1, 35)) +

```

```
labs(x = NULL, y = "Tertiary Education") + theme_minimal(base_size = 12) +
  theme(axis.text.y = element_blank(), axis.ticks.y = element_blank(),
        panel.grid.major.y = element_blank(), panel.grid.minor =
element_blank(), legend.position = "none")
```

Generate Difference by Educ

```
#Non-Tertiary - Tertiary
diff_df <- ame_all %>%
  select(type_label, is_tertiary, AME) %>%
  pivot_wider(names_from = is_tertiary, values_from = AME, names_prefix =
"tert_") %>%
  mutate(
    diff = tert_0 - tert_1,
    diff_label = sprintf("%+.0f", round(diff))
  )

p_diff <- ggplot(diff_df, aes(x = type_label, y = 0)) +
  geom_text(aes(label = diff_label), size = 4, fontface = "bold", color =
"grey30", hjust = 0) +
  coord_flip(ylim = c(0, 1)) +
  labs(x = NULL, y = "AME\nDifference") +
  theme_minimal(base_size = 12) +
  theme(
    axis.text = element_blank(),
    axis.ticks = element_blank(),
    panel.grid = element_blank(),
    axis.title.y = element_blank(),
    axis.title.x = element_text(color = "grey30", size = 9, face =
"bold.italic")
  )
```

Compile and Save

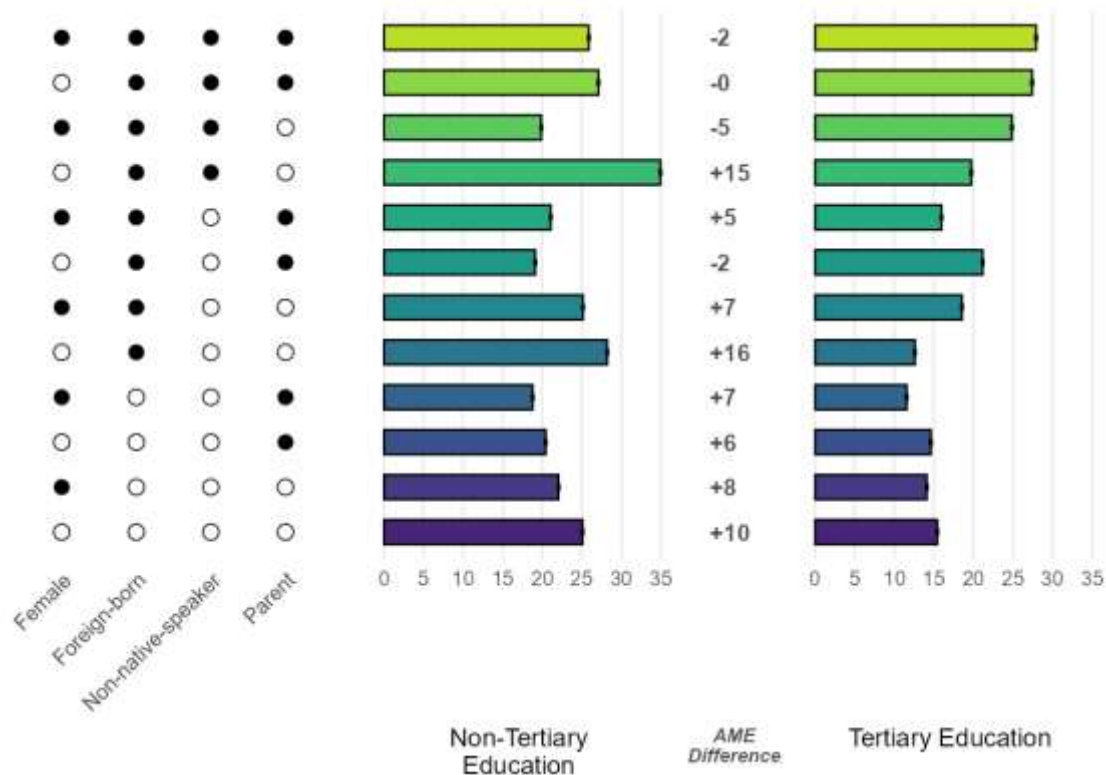
```
agg_png(here("Results", "Fig3.png"), res = 144, height = 750, width = 1060)

ggarrange(
  p_dots,
  p_bar,
  p_diff,
  p_bar_t,
  ncol = 4,
  widths = c(1.4, 1.4, 0.35, 1.4),
  align = "h"
)

dev.off()

## png
## 2

include_graphics(here("Results", "Fig3.png"))
```



Shiny App Precompute

The RAM required is very high, and therefore impractical to run in Shiny. Therefore, precompute here and upload.

Country gains function

```
calc_country_gains <- function(df2_full, spec, AME, ctx_data,
weight_var="aweight") {

  df <- df2_full %>%
    left_join(ctx_data, by = "iso3c") %>%
    mutate(context_effect = replace_na(context_effect, 0))

  g <- in_group(df, spec)
  g[is.na(g)] <- FALSE

  treatable <- g & df$NFE12 == 0 & !is.na(df$skill_reading)
  treatable[is.na(treatable)] <- FALSE

  df <- df %>%
    mutate(
      # With Country slope Context Effect
      net_effect_ctx = pmax(0, AME + context_effect),
      skill_boost_ctx = ifelse(treatable, skill_reading + net_effect_ctx,
skill_reading),
```

```

    # Baseline AME Only
    net_effect_base = pmax(0, AME),
    skill_boost_base = ifelse(treatable, skill_reading + net_effect_base,
skill_reading),

    g_total      = ifelse(g, 1, 0),
    g_noale      = ifelse(g & NFE12 == 0, 1, 0)
  )

  df$g_noale[is.na(df$g_noale)] <- 0

  out <- df %>%
    filter(!is.na(iso3c)) %>%
    group_by(iso3c) %>%
    summarise(
      skill_base = weighted.mean(skill_reading, .data[[weight_var]], na.rm =
TRUE),

      # Export both versions
      skill_gain_ctx = weighted.mean(skill_boost_ctx, .data[[weight_var]],
na.rm = TRUE) - skill_base,
      skill_gain_base = weighted.mean(skill_boost_base, .data[[weight_var]],
na.rm = TRUE) - skill_base,

      pct_group      = weighted.mean(g_total, .data[[weight_var]], na.rm =
TRUE),
      pct_group_noale = weighted.mean(g_noale, .data[[weight_var]], na.rm =
TRUE),
      group_skill_base = weighted.mean(skill_reading[g],
.data[[weight_var]][g], na.rm = TRUE),
      pop18to65 = sum(SPFWT0, na.rm = TRUE),
      context_effect = first(context_effect),
      .groups = "drop"
    )

  out
}

```

Shiny App Pre-computation (5-Way Intercategorical)

```

# Context Effects (m5) from the Pooled Model
df2r$m4_baseline_pred <- predict(m4_pooled, re.form = NA)
m5_pooled <- lmer(skill_reading ~ 0 + offset(m4_baseline_pred) + (1 +
NFE12_within | iso3c), data = df2r)

ctx_df <- ranef(m5_pooled)$iso3c %>%
  rownames_to_column("iso3c") %>%
  select(iso3c, context_effect = NFE12_within)

```

```

# Define 5-Way
vars_raw <- c("female", "FORBORN", "FORLANG", "children", "ed3_tertiary")
map_within <- c(female="female_within", FORBORN="forborn_within",
FORLANG="forlang_within", children="children_within",
ed3_tertiary="ed3_tertiary_within")
map_between <- c(female="female_between", FORBORN="forborn_between",
FORLANG="forlang_between", children="children_between",
ed3_tertiary="ed3_tertiary_between")

spec_grid <- expand.grid(
  female      = c(NA, 0L, 1L),
  FORBORN     = c(NA, 0L, 1L),
  FORLANG     = c(NA, 0L, 1L),
  children    = c(NA, 0L, 1L),
  ed3_tertiary = c(NA, 0L, 1L)
)

make_spec_id <- function(row) {
  paste0(
    "f=", ifelse(is.na(row$female), "all", row$female), ";",
    "b=", ifelse(is.na(row$FORBORN), "all", row$FORBORN), ";",
    "l=", ifelse(is.na(row$FORLANG), "all", row$FORLANG), ";",
    "c=", ifelse(is.na(row$children), "all", row$children), ";",
    "e=", ifelse(is.na(row$ed3_tertiary), "all", row$ed3_tertiary)
  )
}

spec_grid$spec_id <- apply(spec_grid, 1, function(x)
make_spec_id(as.list(x)))

# Helpers
make_newdata_for_pooled <- function(df_base, spec, nfe_value) {
  nd <- df_base
  nd$NFE12_within <- nfe_value - nd$NFE12_between
  for (v in vars_raw) {
    target <- spec[[v]]
    if (!is.na(target)) nd[[map_within[[v]]]] <- target -
nd[[map_between[[v]]]]
  }
  nd
}

calc_ame_pooled <- function(model, df_base, spec, weight_var="aweight") {
  nd0 <- make_newdata_for_pooled(df_base, spec, 0)
  nd1 <- make_newdata_for_pooled(df_base, spec, 1)
  p0 <- predict(model, newdata = nd0, re.form = NA)
  p1 <- predict(model, newdata = nd1, re.form = NA)

  d <- p1 - p0
  w <- df_base[[weight_var]]
  ok <- is.finite(d) & is.finite(w) & !is.na(w) & w > 0
}

```

```

d <- d[ok]; w <- w[ok]
ame <- weighted.mean(d, w)
se <- sqrt(sum(w^2 * (d - ame)^2) / (sum(w)^2))
list(AME = ame, SE = se, lower = ame - 1.96*se, upper = ame + 1.96*se)
}

in_group <- function(df, spec) {
  idx <- rep(TRUE, nrow(df))
  for (v in vars_raw) {
    target <- spec[[v]]
    if (!is.na(target)) idx <- idx & (df[[v]] == target)
  }
  idx
}

```

Main computation

```

precomp_combined <- pmap_dfr(spec_grid, function(female, FORBORN, FORLANG,
children, ed3_tertiary, spec_id) {
  spec <- list(female = female, FORBORN = FORBORN, FORLANG = FORLANG,
children = children, ed3_tertiary = ed3_tertiary)

  ame <- calc_ame_pooled(m4_pooled, df2r, spec)
  cg <- calc_country_gains(df2_sub, spec, AME = ame$AME, ctx_data = ctx_df,
weight_var = "aweight")

  cg$spec_id <- spec_id
  cg$AME <- ame$AME
  cg$SE <- ame$SE
  cg$lower <- ame$lower
  cg$upper <- ame$upper
  cg
})

# Save for deployment in Shiny
saveRDS(precomp_combined, here("Shiny", "precomp_spec_country.rds"))
saveRDS(spec_grid, here("Shiny", "precomp_spec_grid.rds"))

```

Citations

```

packages %>%
  map(citation) %>%
  print(style = "text")

## [[1]]
## Wickham H, Averick M, Bryan J, Chang W, McGowan LD, François R,
## Grolemund G, Hayes A, Henry L, Hester J, Kuhn M, Pedersen TL, Miller E,
## Bache SM, Müller K, Ooms J, Robinson D, Seidel DP, Spinu V, Takahashi
## K, Vaughan D, Wilke C, Woo K, Yutani H (2019). "Welcome to the
## tidyverse." Journal of Open Source Software, 4(43), 1686.
## doi:10.21105/joss.01686 <https://doi.org/10.21105/joss.01686>.

```

```

##
## [[2]]
## Wickham H, Hester J, Bryan J (2024). _readr: Read Rectangular Text
## Data_. R package version 2.1.5,
## <https://CRAN.R-project.org/package=readr>.
##
## [[3]]
## Xie Y (2025). _knitr: A General-Purpose Package for Dynamic Report
## Generation in R_. R package version 1.50, <https://yihui.org/knitr/>.
##
## Xie Y (2015). _Dynamic Documents with R and knitr_, 2nd edition.
## Chapman and Hall/CRC, Boca Raton, Florida. ISBN 978-1498716963,
## <https://yihui.org/knitr/>.
##
## Xie Y (2014). "knitr: A Comprehensive Tool for Reproducible Research in
## R." In Stodden V, Leisch F, Peng RD (eds.), _Implementing Reproducible
## Computational Research_. Chapman and Hall/CRC. ISBN 978-1466561595.
##
## [[4]]
## Müller K (2020). _here: A Simpler Way to Find Your Files_. R package
## version 1.0.1, <https://CRAN.R-project.org/package=here>.
##
## [[5]]
## Wickham H (2016). _ggplot2: Elegant Graphics for Data Analysis_.
## Springer-Verlag New York. ISBN 978-3-319-24277-4,
## <https://ggplot2.tidyverse.org>.
##
## [[6]]
## Slowikowski K (2024). _ggrepel: Automatically Position Non-Overlapping
## Text Labels with 'ggplot2'_. R package version 0.9.6,
## <https://CRAN.R-project.org/package=ggrepel>.
##
## [[7]]
## Pedersen T, Shemanarev M (2024). _ragg: Graphic Devices Based on AGG_.
## R package version 1.3.3, <https://CRAN.R-project.org/package=ragg>.
##
## [[8]]
## Zhu H (2024). _kableExtra: Construct Complex Table with 'kable' and
## Pipe Syntax_. R package version 1.4.0,
## <https://CRAN.R-project.org/package=kableExtra>.
##
## [[9]]
## Arel-Bundock V, Enevoldsen N, Yetman C (2018). "countrycode: An R
## package to convert country names and country codes." _Journal of Open
## Source Software_, *3*(28), 848. <https://doi.org/10.21105/joss.00848>.
##
## [[10]]
## Wickham H, Bryan J (2025). _readxl: Read Excel Files_. R package
## version 1.4.5, <https://CRAN.R-project.org/package=readxl>.
##

```

```
## [[11]]
## Long JA (2022). _jtools: Analysis and Presentation of Social Scientific
## Data_. R package version 2.2.0,
## <https://cran.r-project.org/package=jtools>.
##
## [[12]]
## Bates D, Mächler M, Bolker B, Walker S (2015). "Fitting Linear
## Mixed-Effects Models Using lme4." _Journal of Statistical Software_,
## *67*(1), 1-48. doi:10.18637/jss.v067.i01
## <https://doi.org/10.18637/jss.v067.i01>.
##
## [[13]]
## Kuznetsova A, Brockhoff PB, Christensen RHB (2017). "lmerTest Package:
## Tests in Linear Mixed Effects Models." _Journal of Statistical
## Software_, *82*(13), 1-26. doi:10.18637/jss.v082.i13
## <https://doi.org/10.18637/jss.v082.i13>.
##
## [[14]]
## Bolker B, Robinson D (2024). _broom.mixed: Tidying Methods for Mixed
## Models_. R package version 0.2.9.6,
## <https://CRAN.R-project.org/package=broom.mixed>.
##
## [[15]]
## Garnier, Simon, Ross, Noam, Rudis, Robert, Camargo, Pedro A, Sciaini,
## Marco, Scherer, Cédric (2024). _viridis(Lite) - Colorblind-Friendly
## Color Maps for R_. doi:10.5281/zenodo.4679423
## <https://doi.org/10.5281/zenodo.4679423>, viridis package version
## 0.6.5, <https://sjmgarnier.github.io/viridis/>.
##
## [[16]]
## Lumley T (2024). "survey: analysis of complex survey samples." R
## package version 4.4.
##
## Lumley T (2004). "Analysis of Complex Survey Samples." _Journal of
## Statistical Software_, *9*(1), 1-19. R package version 2.2.
##
## Lumley T (2010). _Complex Surveys: A Guide to Analysis Using R: A Guide
## to Analysis Using R_. John Wiley and Sons.
##
## [[17]]
## Kassambara A (2023). _ggpubr: 'ggplot2' Based Publication Ready Plots_.
## R package version 0.6.0, <https://CRAN.R-project.org/package=ggpubr>.
##
## [[18]]
## Wickham H, Henry L (2025). _purrr: Functional Programming Tools_. R
## package version 1.0.4, <https://CRAN.R-project.org/package=purrr>.
```

Colophon

`sessionInfo()`

```

## R version 4.4.1 (2024-06-14 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] grid      stats      graphics  grDevices utils      datasets  methods
## [8] base
##
## other attached packages:
## [1] ggpubr_0.6.0      survey_4.4-2      survival_3.6-4
## [4] viridis_0.6.5     viridisLite_0.4.2 broom.mixed_0.2.9.6
## [7] lmerTest_3.1-3     lme4_1.1-37       Matrix_1.7-0
## [10] jtools_2.3.0       readxl_1.4.5      countrycode_1.6.0
## [13] kableExtra_1.4.0   ragg_1.3.3        ggrepel_0.9.6
## [16] here_1.0.1         knitr_1.50        lubridate_1.9.4
## [19] forcats_1.0.0      stringr_1.5.1     dplyr_1.1.4
## [22] purrr_1.0.4        readr_2.1.5       tidyr_1.3.1
## [25] tibble_3.2.1       ggplot2_3.5.1     tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] Rdpack_2.6.3      DBI_1.2.3          gridExtra_2.3
## [4] rlang_1.1.5       magrittr_2.0.3     furrr_0.3.1
## [7] compiler_4.4.1    png_0.1-8          systemfonts_1.2.1
## [10] vctrs_0.6.5       pkgconfig_2.0.3    crayon_1.5.3
## [13] fastmap_1.2.0     backports_1.5.0    labeling_0.4.3
## [16] pander_0.6.6      rmarkdown_2.29     tzdb_0.5.0
## [19] nloptr_2.2.1      xfun_0.51          uuid_1.2-1
## [22] broom_1.0.7       parallel_4.4.1     R6_2.6.1
## [25] stringi_1.8.4     parallelly_1.45.0  car_3.1-3
## [28] boot_1.3-30       cellranger_1.1.0   numDeriv_2016.8-1.1
## [31] Rcpp_1.0.14       assertthat_0.2.1   pacman_0.5.1
## [34] splines_4.4.1     timechange_0.3.0   tidyselect_1.2.1
## [37] rstudioapi_0.17.1 abind_1.4-8        yaml_2.3.10
## [40] codetools_0.2-20  listenv_0.9.1      lattice_0.22-6
## [43] withr_3.0.2       flextable_0.9.7    askpass_1.2.1
## [46] evaluate_1.0.3    future_1.58.0      huxtable_5.6.0
## [49] zip_2.3.2         xml2_1.3.8         pillar_1.10.1

```

## [52] carData_3.0-5	reformulas_0.4.0	generics_0.1.3
## [55] rprojroot_2.0.4	hms_1.1.3	munsell_0.5.1
## [58] scales_1.3.0	minqa_1.2.8	globals_0.18.0
## [61] glue_1.8.0	gdtools_0.4.2	tools_4.4.1
## [64] data.table_1.17.0	ggsignif_0.6.4	cowplot_1.1.3
## [67] mitools_2.4	rbibutils_2.3	colorspace_2.1-1
## [70] nlme_3.1-164	Formula_1.2-5	cli_3.6.4
## [73] textshaping_1.0.0	officer_0.6.8	
fontBitstreamVera_0.1.1		
## [76] svglite_2.1.3	gtable_0.3.6	rstatix_0.7.2
## [79] fontquiver_0.2.1	digest_0.6.37	farver_2.1.2
## [82] htmltools_0.5.8.1	lifecycle_1.0.4	fontLiberation_0.1.0
## [85] openssl_2.3.2	MASS_7.3-60.2	